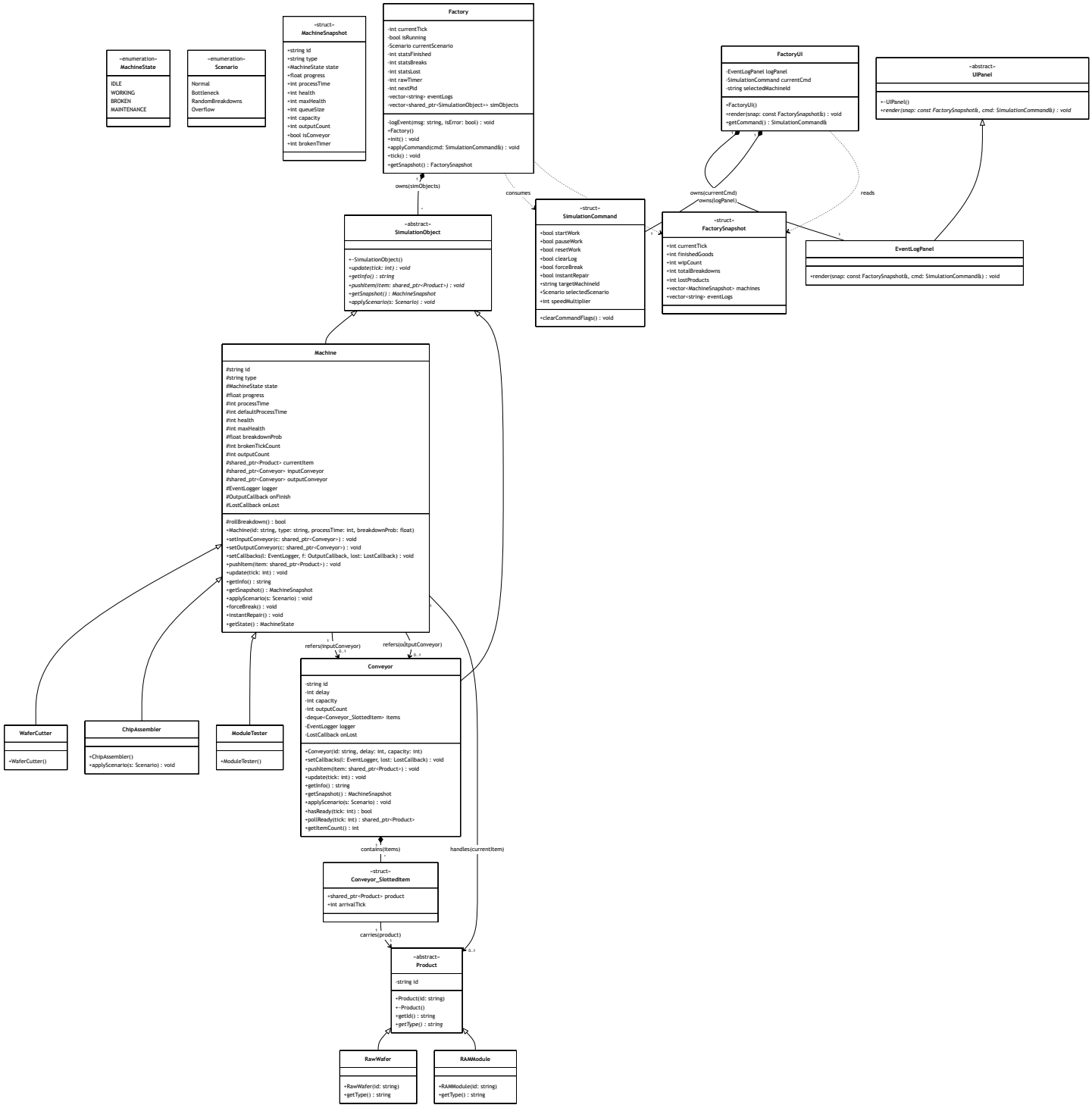
«enumeration»
MachineState
IDLE
WORKING
BROKEN
MAINTENANCE
«enumeration»
Scenario
Normal
Bottleneck
RandomBreakdowns
Overflow
«struct»
MachineSnapshot
+string id
+string type
+MachineState state
+float progress
+int processTime
+int health
+int maxHealth
+int queueSize
+int capacity
+int outputCount
+bool isConveyor
+int brokenTimer
Factory
-int currentTick
-bool isRunning
-Scenario currentScenario
-int statsFinished
-int statsBreaks
-int statsLost
-int rawTimer
-int nextPid
-vector<string> eventLogs
-vector<shared_ptr<SimulationObject>> simObjects
-logEvent(msg: string, isError: bool) : void
+Factory()
+init() : void
+applyCommand(cmd: SimulationCommand&) : void
+tick() : void
+getSnapshot() : FactorySnapshot
FactoryUI
-EventLogPanel logPanel
-SimulationCommand currentCmd
-string selectedMachineId
+FactoryUI()
+render(snap: const FactorySnapshot&) : void
+getCommand() : SimulationCommand&
«abstract»
UIPanel
+~UIPanel()
+render(snap: const FactorySnapshot&, cmd: SimulationCommand&) : void
owns(simObjects)
consumes
owns(currentCmd)
owns(logPanel)
reads
«abstract»
SimulationObject
+~SimulationObject()
+update(tick: int) : void
+getInfo() : string
+pushItem(item: shared_ptr<Product>) : void
+getSnapshot() : MachineSnapshot
+applyScenario(s: Scenario) : void
«struct»
SimulationCommand
+bool startWork
+bool pauseWork
+bool resetWork
+bool clearLog
+bool forceBreak
+bool instantRepair
+string targetMachineId
+Scenario selectedScenario
+int speedMultiplier
+clearCommandFlags() : void
«struct»
FactorySnapshot
+int currentTick
+int finishedGoods
+int wipCount
+int totalBreakdowns
+int lostProducts
+vector<MachineSnapshot> machines
+vector<string> eventLogs
EventLogPanel
+render(snap: const FactorySnapshot&, cmd: SimulationCommand&) : void
Machine
#string id
#string type
#MachineState state
#float progress
#int processTime
#int defaultProcessTime
#int health
#int maxHealth
#float breakdownProb
#int brokenTickCount
#int outputCount
#shared_ptr<Product> currentItem
#shared_ptr<Conveyor> inputConveyor
#shared_ptr<Conveyor> outputConveyor
#EventLogger logger
#OutputCallback onFinish
#LostCallback onLost
#rollBreakdown() : bool
+Machine(id: string, type: string, processTime: int, breakdownProb: float)
+setInputConveyor(c: shared_ptr<Conveyor>) : void
+setOutputConveyor(c: shared_ptr<Conveyor>) : void
+setCallbacks(l: EventLogger, f: OutputCallback, lost: LostCallback) : void
+pushItem(item: shared_ptr<Product>) : void
+update(tick: int) : void
+getInfo() : string
+getSnapshot() : MachineSnapshot
+applyScenario(s: Scenario) : void
+forceBreak() : void
+InstantRepair() : void
+getState() : MachineState
refers(inputConveyor)
refers(outputConveyor)
Conveyor
-string id
-int delay
-int capacity
-int outputCount
-deque<Conveyor_SlottedItem> items
-EventLogger logger
-LostCallback onLost
+Conveyor(id: string, delay: int, capacity: int)
+setCallbacks(l: EventLogger, lost: LostCallback) : void
+pushItem(item: shared_ptr<Product>) : void
+update(tick: int) : void
+getInfo() : string
+getSnapshot() : MachineSnapshot
+applyScenario(s: Scenario) : void
+hasReady(tick: int) : bool
+pollReady(tick: int) : shared_ptr<Product>
+getItemCount() : int
WaferCutter
+WaferCutter()
ChipAssembler
+ChipAssembler()
+applyScenario(s: Scenario) : void
ModuleTester
+ModuleTester()
contains(items)
handles(currentItem)
«struct»
Conveyor_SlottedItem
+shared_ptr<Product> product
+int arrivalTick
carries(product)
«abstract»
Product
-string id
+Product(id: string)
+~Product()
+getId() : string
+getType() : string
RawWafer
+RawWafer(id: string)
+getType() : string
RAMModule
+RAMModule(id: string)
+getType() : string